

Can LLM Agents Keep Disaster Monitoring Alive?

An Evidence-Grounded Closed-Loop Benchmark for Agentic Communication

Working draft — `adaptive-communication` repository

Abstract—Disaster-agent benchmarks typically expose a heterogeneous data environment and ask an agent to assemble evidence, call tools, and produce an answer. Real pre-disaster monitoring adds a harder constraint: evidence acquisition, context freshness, control actions, and the physical outcome are coupled through the same intermittent communication system. We formulate this setting as *evidence-grounded closed-loop Agentic Communication*. A source-grounded physical/data plane produces a versioned Evidence World; an externally authorised Operational Task is compiled into a typed Agent runtime with explicit EvidenceNeed, Context, and Capability contracts; and device actions execute on the same battery, LoRaWAN Class A, backhaul, backup, and direct-to-satellite simulator used for communication scoring. The benchmark separates model diagnostics from physical outcome through R0–R3 replay, owner-scoped evidence semantics, action lifecycle tracking, strong deterministic baselines, evaluator-only oracles, and cumulative gold replacement. Current deterministic validation establishes runtime/physics equivalence, exposes the cost of eager diagnosis and fixed-order probing, validates source-period and robustness coordinates, and verifies layer-wise attribution without fabricating model scores. The remaining model study uses frozen task-conditioned, FullDump, and generic-ReAct inputs under the same capability surface and physical reference.

Index Terms—Agentic communication, disaster monitoring, LLM agents, intermittent connectivity, evidence, context, LoRaWAN, resilient sensing, benchmark.

I. INTRODUCTION

Recent disaster-agent benchmarks demonstrate that language-model agents can orchestrate heterogeneous geospatial data and typed tools across long workflows [1]. Their world, however, is primarily a readable computational environment: once a dataset or tool endpoint is available, the agent can invoke it without changing the underlying disaster-monitoring network. That abstraction omits a central difficulty of field operations. In a mountain monitoring system, the evidence an agent needs may reside at a node or gateway, the path needed to retrieve it may be the path that has failed, and the same scarce communication opportunity may also be needed to install the corrective action.

We study this closed loop in pre-disaster geohazard monitoring. Battery/solar sensor nodes report through LoRaWAN Class A to a field gateway; cellular backhaul can fail intermittently; backup, terminal direct-to-satellite (DtS), access assist, and remote sampling/report configuration are available communication capabilities. Hazard authority remains external: the

Agent does not predict a landslide or invent a warning level. Its job is to execute an authorised monitoring task while evidence, context, and actuation are constrained by the communication system itself.

The paper is organised around two method objects:

$$\boxed{\text{Physical/Data Plane} \rightarrow \text{Evidence World}} \quad (1)$$

and

$$\boxed{\text{Operational Task} \rightarrow \text{Task-conditioned Agent Runtime}} \quad (2)$$

The first gives observations explicit provenance, owner, revision, freshness, reachability and status semantics. The second turns a benchmark-level monitoring task into Runtime TaskContracts, EvidenceNeeds, versioned Context, typed evidence-use/device-use capabilities, structured planner decisions, and physical effects. Both are evaluated on the same simulator and scorer as ordinary communication baselines.

A. Contributions and current boundary

The current artifact makes four concrete contributions.

- 1) **A closed-loop benchmark abstraction.** We distinguish Operational Task semantics from Agent-runtime TaskContracts, and define a typed Evidence World that never exposes simulator hidden truth as Agent evidence.
- 2) **A unified capability/runtime interface.** Evidence reads and communication-device actions share authority, owner/path, failure, lifecycle, latency/resource, and effect contracts; five device capabilities execute through existing physical mechanisms.
- 3) **Communication $\times$ Agent evaluation.** Physical monitoring outcome remains primary while R0–R3 replay, frozen-input trajectory metrics, model-call ledgers, oracles, and gold replacement localise failures without collapsing them into an arbitrary overall score.
- 4) **A reproducible experimental substrate.** O1–O6 tasks, source-period splits, robustness coordinates, deterministic Agent baselines, source-derived task transfer, and all paper tables are generated from frozen manifests/results.

No real LLM score is claimed in the current snapshot: the model protocol and transport are frozen, while model runs require an external endpoint/key and cannot silently fall back to a scripted consumer.

II. SCENARIO AND SYSTEM MODEL

We consider pre-disaster mountain monitoring with nodes $i \in \mathcal{N}$, a gateway g , and a centre c . Time is slotted. A monitoring obligation

$$o = (i, m, r_o, [s_o, e_o], d_o) \quad (3)$$

requires measurand m from node i to be sampled within $[s_o, e_o]$ and delivered centrally by d_o . Its success indicator is I_o^{succ} and the primary service metric is

$$\text{TDR} = \frac{1}{|\mathcal{O}|} \sum_{o \in \mathcal{O}} I_o^{\text{succ}}. \quad (4)$$

Collection, missing delivery, censoring, latency, Age of Information (AoI), recovery and resource metrics are reported separately; the Agent never removes an obligation from the denominator.

Each node maintains battery $B_{i,t}$, source cache $Q_{i,t}$ and installed configuration $C_{i,t}$. Battery evolution follows

$$B_{i,t+1} = \min \left\{ B_i^{\text{max}}, [B_{i,t} + H_{i,t} - E_{i,t}^{\text{sense}} - E_{i,t}^{\text{UL}} - E_{i,t}^{\text{DL}} - E_{i,t}^{\text{DtS}} - E_{i,t}^{\text{other}}]^+ \right\}, \quad (5)$$

where $H_{i,t}$ can be driven by source-period-separated NASA POWER irradiance. A successful Class A uplink opens bounded receive opportunities; a centre-originated configuration therefore requires both a return/control path and a valid access opportunity. Gateway backup, terminal DtS, and access-assist reuse their existing physical implementations and counters rather than Agent-reported success.

The complete physical state is x_t and exogenous process is ξ_t :

$$x_{t+1} = F(x_t, a_t, \xi_t; \phi). \quad (6)$$

The Agent does not observe x_t . At owner/location ℓ , only a legal observation $z_t = H(x_{\leq t}, a_{< t}, \xi_{\leq t}, \ell)$ can enter the runtime.

III. DATA AND EVIDENCE WORLD

A typed evidence item is

$$e_j = (p_j, v_j, s_j, o_j, t_j^{\text{gen}}, t_j^{\text{obs}}, \rho_j, \nu_j, \sigma_j), \quad (7)$$

containing proposition/value, source/provenance, owner, generation/observation times, revision, freshness semantics and status. Status distinguishes at least CURRENT, STALE, NONE_RECENT, UNREACHABLE, TIMEOUT and NEGATIVE observation. These distinctions prevent an absent packet from being silently interpreted as a negative fact.

The Evidence World at time t is

$$\mathcal{E}_t = \{e_j : t_j^{\text{obs}} \leq t, e_j \text{ passes the runtime gate}\}. \quad (8)$$

Evidence revision changes only when factual content changes; age and AoI are derived at materialisation time. Owner-scoped gateway reads cross the same backhaul abstraction: when that path is unavailable the Agent receives UNREACHABLE rather than gateway truth.

The primary benchmark uses a source-grounded hybrid simulator: SRTM/ITM terrain propagation, NASA POWER irradiance/temperature, ChirpBox-derived temporal link processes,

public communication and geohazard-monitoring constraints, and the frozen event-driven monitoring simulator. Source years are split before random seeds; the current manifest uses 2022/2023/2024 as train/dev/test source periods. Hazard physics may later generate Operational Tasks, but landslide prediction is outside the Agent control loop.

IV. AGENT ABSTRACTION AND RUNTIME

We separate two meanings of “task.” An *Operational Task* is the benchmark/business objective: target nodes/measurands, monitoring profile, windows/deadlines, authorised effects and scoring profile. A *Runtime TaskContract* is an Agent program object with desired state, evidence contract, temporal contract, effect ceiling and completion predicate.

For a runtime task T_j , the ContextManifest is

$$C_t = M(T_j, S_t, \mathcal{E}_t, K_t), \quad (9)$$

where S_t is InvestigationState and K_t is the visible capability catalogue. The model input is $P_t = \text{Materialize}(C_t)$; a planner emits

$$(q_t, a_t, \Delta S_t, \text{stop}_t) = \pi(T_j, P_t, K_t). \quad (10)$$

Capabilities split into evidence-use $\mathcal{K}^E$ and device-action $\mathcal{K}^A$ but share one contract surface. An action lifecycle distinguishes requested, accepted/refused, delivered, applied and confirmed. Hence lack of confirmation never implies that an action was not applied.

The live runtime supports multi-round execution:

$$C_t^{(k)} \rightarrow q_t^{(k)} \rightarrow \mathcal{E}_t^{(k+1)} \rightarrow C_t^{(k+1)} \rightarrow a_t. \quad (11)$$

In particular, a planner can request gateway-owner evidence, trigger an Evidence World revision, then decide whether to configure a node or invoke a communication fallback.

V. BENCHMARK TASKS AND EXPERIMENTAL PROTOCOL

The benchmark contains six Operational Task families: O1 monitoring continuity, O2 risk escalation, O3 backhaul-outage sustainment, O4 energy-constrained monitoring, O5 recovery and reconciliation, and O6 compound long-horizon operation. The same deployment, physical transition, scorer and information boundary are reused across methods.

We use four replay levels. R0 replays protocol/state transitions; R1 substitutes a planner on frozen model inputs; R2 reconstructs exact Context materialisation; R3 reruns the full simulator. The model-facing protocol records ModelRequest, ModelAttempt, ModelUsage and PlannerDecision. R1 diagnostics include stopping correctness, tool selection precision/recall, order LCS and argument grounding. R3 reports their physical communication consequence.

Online baselines and evaluator-only oracles are typed separately. Current online references include local/ordinary communication control, deterministic task comply, task-conditioned evidence, diagnosis-first, fixed-order eager probing, FullDump, and generic-ReAct context. Delivery/energy oracles are evaluator-only and never appear as online Agent arms. A generic VoI baseline is intentionally deferred until evidence-acquisition cost/value has a source-backed or

TABLE I

O2 GLOBAL RUNTIME CONFORMANCE. ALL VALUES ARE GENERATED FROM FROZEN RESULT JSON.

Arm	TDR (%)	Collection (%)	p90 latency (s)	Energy (Wh)	Backup (B)	Physical = ref
legacy comply	32.44	51.06	468.0	0.35077	2141.6	yes
task-conditioned runtime	32.44	51.06	468.0	0.35077	2141.6	yes
FullDump runtime	32.44	51.06	468.0	0.35077	2141.6	yes

TABLE II

LOCALISED O2 CONTEXT MATERIALISATION UNDER PAIRED PHYSICAL EXECUTION.

Context	Selected evidence	Materialized bytes	Sufficiency	Action confirm (%)	Physical = ref
task-conditioned	10.32	19763.6	1.000	85.00	yes
FullDump	27.32	29575.6	1.000	85.00	yes

simulator-authoritative scale; arbitrary weights would manufacture a comparison rather than strengthen it.

Gold replacement follows the cumulative order

$$\begin{aligned}
 &\text{Task} \rightarrow \text{EvidenceNeed} \rightarrow \text{Percept} \rightarrow \text{Context} \\
 &\quad \rightarrow \text{Selection} \rightarrow \text{Order} \rightarrow \text{Arguments} \\
 &\quad \rightarrow \text{Policy} \rightarrow \text{Physical Oracle}.
 \end{aligned} \tag{12}$$

Runtime-owned upstream replacements require a model re-run; planner-owned selection/order/argument layers are frozen-input diagnostics; policy/oracle replacements return to R3.

VI. CURRENT DETERMINISTIC VALIDATION

This section reports infrastructure and deterministic evidence only. It is intentionally separate from the future real-model evaluation.

A. Runtime/physics conformance

The self-contained runtime, task-conditioned context and FullDump materialiser reproduce the paired deterministic communication reference while emitting typed Evidence/Context/Capability traces. The global O2 physical summary is generated in Table I.

For a localised O2 scope, task-conditioned materialisation selects fewer irrelevant records while maintaining Context sufficiency and the same physical result (Table II).

B. Deterministic Agent/runtime baselines

Table III holds the physical policy fixed and varies the runtime/context strategy. Diagnosis-first and fixed-order eager probing add tool/model/context work without changing physical outcome. Generic-ReAct context removes explicit EvidenceNeed/InvestigationState artifacts and therefore isolates their model-input footprint without changing the tool surface.

C. Robustness coordinates and task transfer

The current source-period gate reaches the physical energy model rather than only changing a manifest. A broader five-axis gate checks weather, backhaul outage, task scope, evidence owner and deployment scale; these are infrastructure validations, not claims of method robustness.

TABLE III

PAIRED O2 RUNTIME OVERHEAD RELATIVE TO DIRECT DETERMINISTIC COMPLY.

Baseline	Extra model calls	Extra tool calls	Extra contexts	Extra materialized bytes	Physical = ref
task-conditioned evidence	0.0	0.0	0.0	0.0	yes
diagnosis-first	2.0	4.0	2.0	65046.8	yes
fixed-order eager	14.0	28.0	14.0	480537.8	yes
generic ReAct context	0.0	0.0	0.0	-377527.4	yes

TABLE IV

SOURCE-PERIOD SEPARATION REACHES THE PHYSICAL SIMULATOR.

Split	NASA POWER year	Harvested (Wh)	Consumed (Wh)	TDR (%)	Physical = paired ref
train	2022	0.14604	0.03952	94.64	yes
dev	2023	0.14674	0.03952	94.64	yes
test	2024	0.09480	0.03670	94.64	yes

The first source-derived task-transfer experiment maps a published monitoring-stage sequence into the external-authority Operational Task while keeping runtime, planner, capability surface and scorer fixed (Table VI).

D. Attribution protocol self-check

Before applying gold replacement to a real model, we inject controlled upstream and planner corruptions into frozen O2 R1 turns. Table VII verifies that upstream replacement restores the gold assembly, selection repair does not smuggle gold arguments, and argument grounding reaches the gold value only at the argument layer. This table validates the evaluator; it is not a model result.

VII. REAL-MODEL EVALUATION PLAN

The model study uses frozen task-conditioned, FullDump and generic-ReAct inputs that share the same O2 task, capability catalogue, simulator and paired physical reference. R1 first measures planner diagnostics on identical frozen turns. Only model/context combinations passing the declared validity and budget gates proceed to R3, where communication outcome, model/tool calls, latency and physical resources are measured jointly. No result is reported here until a real backend run is present in the frozen result registry.

VIII. RELATED WORK

DORA [1] establishes a valuable benchmark pattern for heterogeneous disaster data, typed tools, expert trajectories and trajectory-level diagnostics. Our benchmark adopts the discipline of typed tools, replayable trajectories and final-primary/trajectory-diagnostic evaluation, while changing the system boundary: evidence is dynamic and owner-scoped, tool access can be unreachable, device actions have delivery/apply/confirm lifecycles, and the final score is a physical monitoring outcome rather than only an answer. Existing LLM-based network planning and emergency-network work [2], [3], [4], [5] motivates the Agent interface; our emphasis is a shared physical communication substrate for evidence acquisition and action.

TABLE V
FIVE-AXIS PAIRED ROBUSTNESS-COORDINATE ACTIVATION.

Axis	Coordinates	Activated signal	Paired physical = ref	Replay exact
weather window	early→late	harvest 0.00000→0.08660 Wh	yes	yes
backhaul outage	short_early→long_late	backup 202→342 B	yes	yes
target scope	global→localized	obligations 1092→498	yes	yes
evidence owner	center_only→center_gateway	center→center/gateway	yes	yes
deployment scale	one_group→two_group	nodes 9→14	yes	yes

TABLE VI
OPERATIONAL TASK TRANSFER UNDER AN UNCHANGED
COMMUNICATION RUNTIME.

Operational Task	Authority	Benchmark phase profile	Paired physical = ref	Replay exact
hand-authored O2	benchmark authority	0h:3600s / 6h:300s	yes	yes
Qili source-derived	S14	0h:3600s / 4h:300s / 8h:3600s	yes	yes

IX. LIMITATIONS AND NEXT EXPERIMENTS

The current artifact has four explicit boundaries. First, real LLM scores are pending an external model endpoint; no scripted backend is substituted. Second, a generic VoI baseline is deferred until observation cost/value admits a defensible scale. Third, current hazard physics remains an optional upstream task generator rather than a claimed landslide predictor. Fourth, the first source-derived transfer validates task translation rather than real-site end-to-end deployment.

Next experiments therefore focus on real model families under the frozen three-context matrix, R1-to-R3 failure propagation through the attribution protocol, additional access-outage/reachable- subset/context-length/owner-count robustness coordinates, and additional source-derived or physics-generated Operational Tasks.

X. CONCLUSION

Agentic Communication becomes materially different from static tool use when the evidence needed to reason and the actions needed to repair the system share the same impaired network. The current artifact closes the benchmark/runtime/evaluation loop needed to study that setting without hidden state, arbitrary task changes or handwritten result tables. Its next claim-bearing step is the real-model experiment, not another change to the underlying communication simulator.

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TABLE VII
CUMULATIVE ATTRIBUTION-PROTOCOL INFRASTRUCTURE VALIDATION.

Cumulative layer	Assembly = gold (%)	Stop correct (%)	Tool exact (%)	Arg acc. (%)	Unresolved arg slots
corrupted base	0.00	85.00	0.00	–	0.00
+ Task	0.00	85.00	0.00	–	0.00
+ EvidenceNeed	0.00	85.00	0.00	–	0.00
+ Percept	0.00	85.00	0.00	–	0.00
+ Context	100.00	85.00	0.00	–	0.00
+ capability selection	100.00	100.00	100.00	0.00	0.85
+ capability order	100.00	100.00	100.00	0.00	0.85
+ capability arguments	100.00	100.00	100.00	100.00	0.00